MAPPA: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom mapping assigns corresponding atoms across a chemical reaction, but the same endpoints can support several assignments, and symmetry can multiply their explicit representations. We introduce MAPPA, a graph-based method that retains alternative fragment placements in compressed form. Fragment growth asks only whether one compatible assignment exists. When growth saturates, a separate operation constructs the distinct compressed placements, which condition the subsequent mapping of the remaining atoms. A cut sweep varies the growth conditions to explore alternative fragment boundaries. After final family aggregation, a separate decoder collects unique fragments, shares local bond-event calculations and composes them under the saved joint constraints. A completed validation on all 1,851 Golden reactions recovers the annotated correspondence among the retained alternatives for 1,830 reactions (98.87%), with all matching, decoding and verification tasks completed and no unresolved verification outcomes. Recovery measures reference inclusion rather than top-ranked prediction accuracy. MAPPA provides an inspectable representation of atom correspondence and bond changes for subsequent chemical assessment.

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1 Introduction

Atom-to-atom mapping identifies corresponding atoms across a chemical reaction. These correspondences support reaction-center identification, template extraction and the preparation of structures for reaction-path calculations. Yet endpoint structures do not always determine a unique assignment. Different correspondences may imply different atom origins and bond changes, while molecular symmetry can produce many

equivalent assignments. A useful mapping method should retain structural alternatives without repeatedly listing permutations of the same choice.

Existing approaches obtain correspondences from learned reaction representations or graph-based optimization. RXNMapper uses attention from a reaction language model (13), LocalMapper learns from chemist-labeled reactions (1), and SLAPMapper combines label refinement with sequential linear assignment (8). For methods returning alternatives, recovering a reference somewhere in the output is a different objective from predicting one correct top-ranked mapping. Symmetry also makes the number of explicit assignments a poor measure of the number of structural choices.

Weighted molecular graphs introduce another distinction. Two assignments can both satisfy a permissive bond-weight matching tolerance while producing different bond events at a finer threshold. Retaining one representative alone can therefore hide information needed for chemical interpretation. The search for correspondences and their interpretation as bond changes should remain separate.

MAPPA separates three decisions: whether a fragment can continue growing, which compressed placements its completed constraints permit, and what bond changes those placements imply. Growth retains one compatible assignment. At saturation, a separate operation constructs the distinct compressed fragment placements, which become conditional branches for mapping the remaining atoms. A cut sweep varies the growth conditions to expose alternatives excluded by greedy fragment commitment. The completed search is reduced to a catalogue of mapping families, and a separate decoder shares local bond-event calculations across its unique fragments. We describe these stages and evaluate reference recovery on the Golden benchmark (9).

2 Method

2.1 Overview and terminology

MAPPA takes two molecular endpoints and returns mapping families, each representing a set of allowed atom correspondences. A *witness* is one explicit assignment. During growth, it establishes that the current fragment has at least one match. A *compressed placement* represents the allowed assignments of a completed fragment. A *search branch* records conditional fragment choices while the remaining atoms are mapped. A final *mapping family* retains the resulting assignments together with the constraints and symmetry relations that connect them.

Figure 1 shows the scientific stages. Matching first explores fragment placements under the requested seed and cut conditions. Finalization and reduction then collect the saved families for each reaction and direction. This catalogue supports two separate operations: local bond-event decoding and reference-membership verification. Neither operation repeats fragment growth.

2.2 Endpoint graphs and matching rules

Let V_R, V_P denote the source and target atom sets, Z_R, Z_P their element labels, and W_R, W_P their symmetric bond-weight matrices. A partial mapping $m : D \subseteq V_R \to V_P$ assigns distinct target atoms to the mapped source atoms and preserves their elements. Explicit hydrogen atoms participate. Coordinates may accompany the endpoints but do not determine fragment growth.

The active source graph contains edges with weight at least b. A growing fragment retains a set A of accepted preservation edges. Its witness must satisfy

$$Z_R(u) = Z_P(m(u)), \qquad W_P(m(u), m(v)) \geq b, \qquad |W_R(u, v) - W_P(m(u), m(v))| \leq \tau_{\text{iso}} \tag{1}$$

for each $(u, v) \in A$, while respecting assignments locked by completed fragments. Only an accepted growth edge adds a preservation constraint. Other bonds between included source atoms are not implicitly required, and source nonedges impose no negative constraints. The full original matrices remain available for subsequent event interpretation.

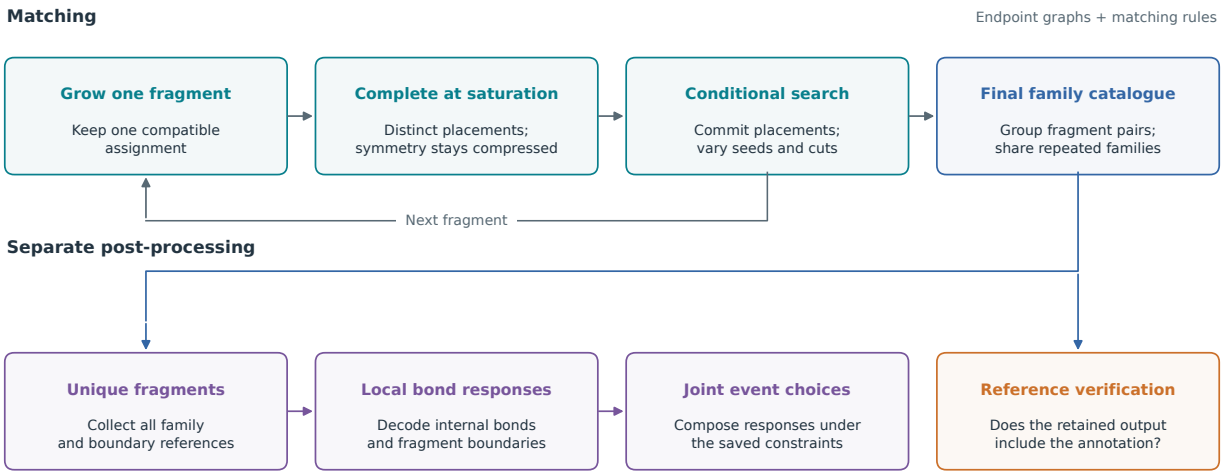


Figure 1 Scientific stages of MAPPA. Growth asks for one compatible assignment; saturation triggers compressed completion and conditional branching. Seed orderings and cut conditions repeat this matching process. Finalized results are reduced to a mapping catalogue before local decoding. Reference verification queries the same catalogue independently of decoding.

2.3 Growing a fragment with one witness

Starting from a seed, MAPPA chooses a boundary atom uniformly and then one of its incident frontier edges uniformly. It proposes adding that edge and asks whether the enlarged fragment has a compatible injective assignment. Bond weight does not rank the proposals.

Whenever possible, the existing witness is extended by checking the new edge and newly joined locked atoms. Otherwise, an exact existence query permits earlier assignments within the growing fragment to change; assignments belonging to previously completed fragments remain fixed. The query removes destinations that cannot satisfy element, bond or distinct-atom requirements, then searches the remaining choices only until it finds one witness. The same query proves absence if all remaining choices are excluded. Detailed propagation and search rules are given in Appendix A.1.

A successful proposal accepts the edge and replaces the witness. Proven absence defers the proposal while other frontier edges are explored. A resource limit remains an unresolved search, rather than a negative match. Joining a completed fragment preserves its locked assignments and accepted constraints. Growth saturates when no further admissible proposal remains. Thus, intermediate growth needs one witness, regardless of how many assignments that fragment could ultimately represent.

2.4 Constructing compressed placements at saturation

At saturation, MAPPA uses the finished fragment constraints to construct its distinct compressed placements. Compression has two complementary forms. An interchangeable *pool* stores target atoms among which every represented injective assignment is valid. A correlated symmetry action describes atoms that must move together. For example, a rotation of a ring changes several atom assignments jointly; allowing its atoms to move independently would introduce invalid combinations.

Completion allocates source atoms to interchangeable target classes and uses the exact symmetry of the globally colored target graph, conditioned on earlier locks and the matching context, to avoid constructing equivalent allocations. Each returned placement carries a witness and the compact choices it represents. Symmetric assignments within that placement remain compressed. Exact graph automorphisms are computed with nauty (10); the class construction and conditioned symmetry action are specified in Appendix A.2.

Local interchangeability and global symmetry serve different purposes. A tolerance can admit several target bond weights without making those weights globally equivalent. Likewise, a symmetry action can move atoms

Algorithm 1 MAPPA matching and catalogue construction

Require: Endpoint graphs, matching policies, seed/cut plan and search budgets
Ensure: A mapping-family catalogue for each search direction
for all requested directions **do**
  Initialize collected search results
  for all cut conditions and seed orderings **do**
    Initialize live branches with any supplied anchors
    while a live branch can continue under the seed ordering **do**
      Select an eligible branch and fragment seed
      Grow with one-witness existence queries until saturation
      if an admissible fragment has been established **then**
        Construct distinct compressed placements of the finished fragment
        Continue each retained placement with completed assignments locked
      else
        Retain the branch for later seeds
      end if
      Enforce hierarchy caps and record completion, stalls or truncation
    end while
    Retain terminal results and their provenance
  end for
  Finalize saved symmetry; group fragment pairs and reduce repeated families
end for

outside the current fragment as well as those inside it. MAPPA retains the joint restrictions needed for later placements. Equal atom sets or individual symmetry labels alone do not justify treating all assignments as interchangeable.

2.5 Conditional branching and cut-sweep exploration

Each retained completed-fragment placement creates one continuation for the remaining atoms. Earlier placements condition later choices, as illustrated in Figure 2a. The branch cap counts compressed choices at each hierarchy level, including the combined live continuations across parents. Assignments represented inside a pool are not separate branches. Expansions exceeding the cap are recorded as truncated; a capped result does not establish exhaustive coverage.

Fragment growth is greedy along a trajectory: accepting an edge can exclude placements that would be available with a different fragment boundary. MAPPA explores alternative trajectories through seed orderings and a cut sweep. For a source graph $G_R = (V_R, E_R)$, the sweep searches the uncut graph and one graph $G_R^{(e)} = (V_R, E_R \setminus \{e\})$ for each eligible edge e. At the evaluated settings, the cut threshold equals the active-graph floor, so the sweep includes $|E_R| + 1$ conditions, including cuts involving H.

Masking an edge changes which preservation constraints growth can acquire. It can redirect growth, expose another placement or leave atoms for a later fragment (Figure 2b). The endpoints of the masked edge may still join through another route, and their final correspondence may preserve the bond. A cut is therefore a search condition, not a prescribed bond event. One cut can redirect several subsequent placements and expose a correspondence with multiple bond changes. Combining both search directions further broadens the explored choices, while retaining each family's orientation. The sweep and seed orderings improve exploration without making greedy search exhaustive.

2.6 Aggregating the final mapping families

After matching, saved symmetry is finalized and results from all supplied seeds and cuts are collected for each reaction and direction. Final branches are grouped by the unordered collection of their matched fragment atom-set pairs,

$$\mathcal{B} = \{(R_1, P_1), \ldots, (R_q, P_q)\}. \tag{2}$$

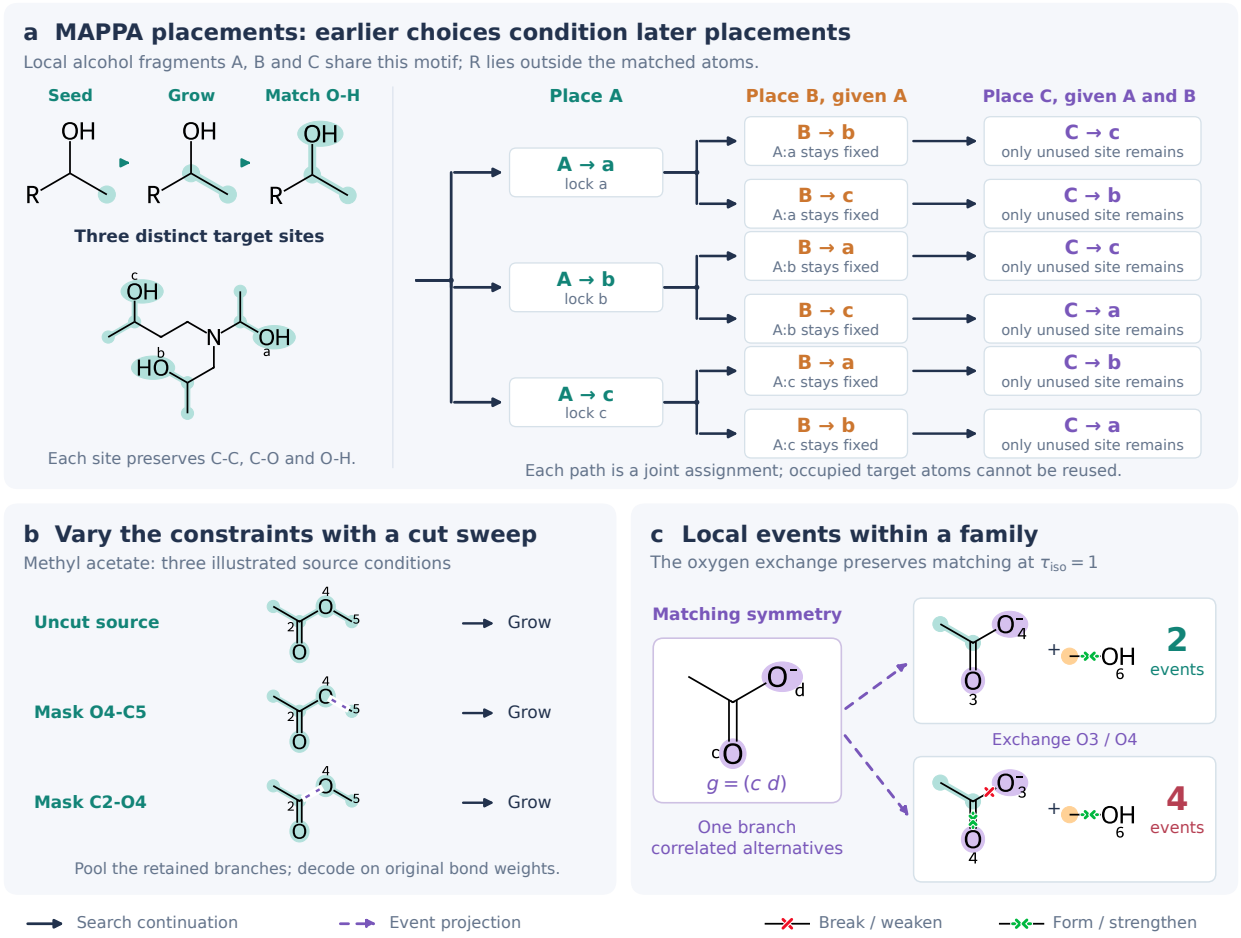


Figure 2 Fragment choices, cut conditions and bond-event interpretation. (a) Three source alcohol motifs can occupy three target sites with distinct molecular environments. Placing A conditions the available sites for B and C. This illustrates injectivity and conditional placement, rather than a full growth trace; attached hydrogens are included in the local matches. (b) Single-edge masks in an ester change the growth conditions, and their search outputs are pooled. A mask alone assigns no bond event. (c) Exchanging acetate oxygen images preserves matching at $\tau_{\mathrm{iso}} = 1$ but changes two heavy-atom events into four at threshold 0.5. Neither acetate oxygen carries H. Black lines show endpoint bonds, red × marks show cleavage or weakening, and green inward arrows show formation or strengthening. Dashed arrows indicate event projections within a family. These formal-bond-order signatures illustrate decoding, not distinct mechanisms or resonance-normalized transformations.

Growing fragment A before B or B before A does not create a new final group when the paired sets agree. Each group nevertheless retains its mapping families: different preserved edges or correlated actions can permit different assignments even when the fragment sets coincide.

Reduction shares identical normalized family records while preserving distinct constraints, alternative fragment partitions and provenance. It preserves the union of represented mappings and does not assume that the remaining families are mutually disjoint. This cross-search reduction is separate from constructing compressed placements during matching. The resulting catalogue is the input to decoding and verification; a single cut is not an independent decoding unit.

2.7 Decoding through shared local bond responses

Decoding translates the saved mapping choices into bond changes using the original endpoint matrices. For mapped atoms u, v, the signed change is

$$\Delta_{uv}(m) = W_P(m(u), m(v)) - W_R(u, v). \tag{3}$$

An event occurs when $|\Delta_{uv}|$ is strictly greater than its threshold: 0.3 for a pair involving a metal and 0.5 otherwise. Positive changes indicate formation or strengthening; negative changes indicate cleavage or weakening. These thresholds are independent of the matching tolerance. Matching-compatible assignments can consequently produce different events (Figure 2c).

The decoder first collects all unique saved fragments across the complete catalogue, together with their family occurrences, constraints, symmetry actions and boundary relationships. Only after this inventory is complete does it calculate reusable local responses. The responses cover bonds within fragments, between fragments and across mapped/unmatched boundaries. Bonds wholly inside unmatched material are omitted; no correspondence is invented for unmatched atoms.

Each bond response depends on the destinations of its two endpoints. If the response is constant over the allowed destinations, it can be recorded directly. Otherwise, the decoder follows the reachable endpoint pairs under the saved actions, avoiding enumeration of the full molecular assignment family. Responses are shared only when their relevant fragment and constraint context agree. Appendix A.5 gives the construction.

For a requested joint choice, the decoder checks the full family constraints and combines the local contributions, counting each bond once. An action shared across several fragments remains one correlated choice. Alternative fragment partitions are alternative descriptions of the same family, so their totals are not added together. Ordinary decoding retains these local responses and composition rules; enumerating whole-reaction event patterns or selecting a minimum-event mapping are separate queries.

2.8 Reference verification and computational organization

Reference verification asks whether an annotated atom relation belongs to the retained mapping catalogue under a declared atom subset and endpoint-equivalence policy. It queries the compressed relation rather than comparing only its representative witness. An unresolved query is reported separately from verified inclusion or exclusion. A reference absent from the saved catalogue may still be compatible with search choices that were not explored.

Matching, family reduction and decoding have separate interfaces for local and distributed execution. Persistent workers reuse completed-fragment results when their recorded dependencies remain valid. The cache stores compressed results and their dependency information; eviction changes computation, not the scientific result. Aggregated storage permits intermediate results to be passed between stages without creating a file for every cut or seed. A decoding task loads one reaction/direction catalogue, prepares its local responses, publishes the result and releases its working data. Technical cache and execution details are given in Appendix A.6.

3 Evaluation

3.1 Golden reference recovery

We evaluated the one-witness/direct-completion design on all 1,851 Golden reactions (9), comprising 3,702 search directions. Inputs use formal bond orders and explicit H, with reference labels removed before search. The run used one seed ordering per cut, random seed 42, branch cap 100, both directions, the full cut sweep, $b = 0.2$ and $\tau_{\text{iso}} = 1$. Event thresholds were 0.5, or 0.3 for pairs involving metals.

Recovery requires the annotated heavy-atom relation, including unmatched atoms, modulo endpoint chemical symmetries preserving charge, isotope, H count and stereochemistry. These attributes define reference equivalence; they do not imply that element/bond-weight matching itself enforces every chemical attribute. A reaction is recovered if either direction contains its reference. This measures inclusion among alternatives, not top-ranked prediction accuracy. Failures and unresolved cases remain in the denominator.

MAPPA recovered 1,830 of 1,851 reactions (98.87%) and 3,618 directional references (Figure 3). Matching, reduction, local decoding and verification completed for every direction, with no failed tasks or unknown verification outcomes. Twenty-one reaction references were absent from the retained families. Their absence is an outcome of the specified bounded search, not a proof that no compatible mapping exists.

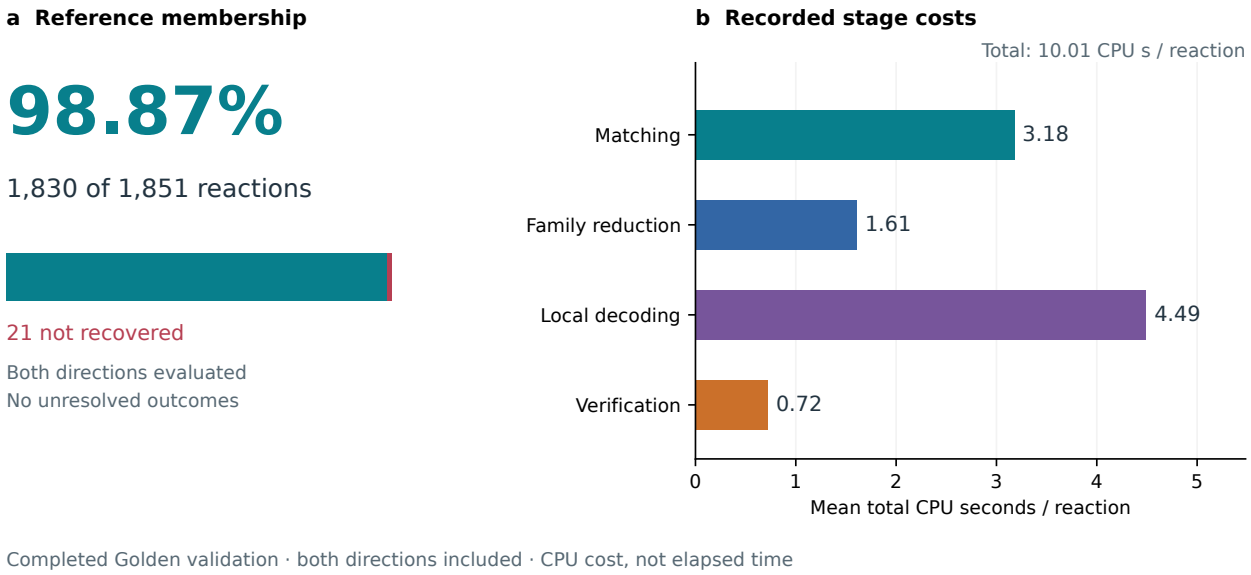


Figure 3 Completed Golden validation of the current algorithmic design. Reference recovery uses all 1,851 reactions and accepts inclusion in either direction. Stage costs are mean total worker CPU seconds per reaction, summed across both directions and including task publication. They are not wall-clock latency. All 3,702 directions completed matching, reduction, local decoding and verification. Source and measurement scope are fixed to the recorded validation run.

These results apply to the recorded completed run. Its source version, per-case outcomes and measurement boundaries are provided in Appendix B; later implementation revisions are not implicitly certified by this evaluation. Historical seed, cap and comparator studies are reported separately in Appendix C.

3.2 Computational cost

Figure 3 separates the measured costs of matching, family reduction, local decoding and reference verification. Summed worker CPU over both directions averaged 10.01 seconds per reaction across all four stages. This includes task computation and publication, but is not elapsed latency or total allocation CPU: coordinator activity, startup and idle capacity have different measurement boundaries. The run completed 177,552 matching tasks. Completed-fragment reuse answered 91.8% of recorded fragment calls; this measures cache reuse, not a causal speedup against an uncached control.

The stage separation exposes both search and interpretation cost. In particular, fast matching does not imply negligible decoding: local response tables still depend on the number of reachable endpoint pairs and retained family contexts. The reported costs characterize this completed run and are not combined with historical timings from different hardware or scientific settings. Full stage measurements and their sampling coverage accompany the per-case evidence.

3.3 Chemical interpretation of retained alternatives

A correspondence describes net atom origins and bond changes without specifying intermediates or elementary steps. Figure 4 illustrates why retaining alternatives can matter. In Golden case 9, the annotated assignment places the alcohol oxygen in the product ring and the ketone oxygen in water. A second assignment exchanges those oxygen origins, while a third also changes two carbon assignments. The second has fewer events than the reference; the first and third have equal event totals but different correspondences. Neither a minimum event count nor a single total identifies the annotated transformation.

The figure uses selected archived witnesses as a chemical illustration. It is not an exhaustive output of the current decoder, and the alternatives are not validated reaction pathways. The historical coordinate/WBO study in Appendix C similarly examines event alternatives under unverified mappings; it is not a mapping-accuracy benchmark.

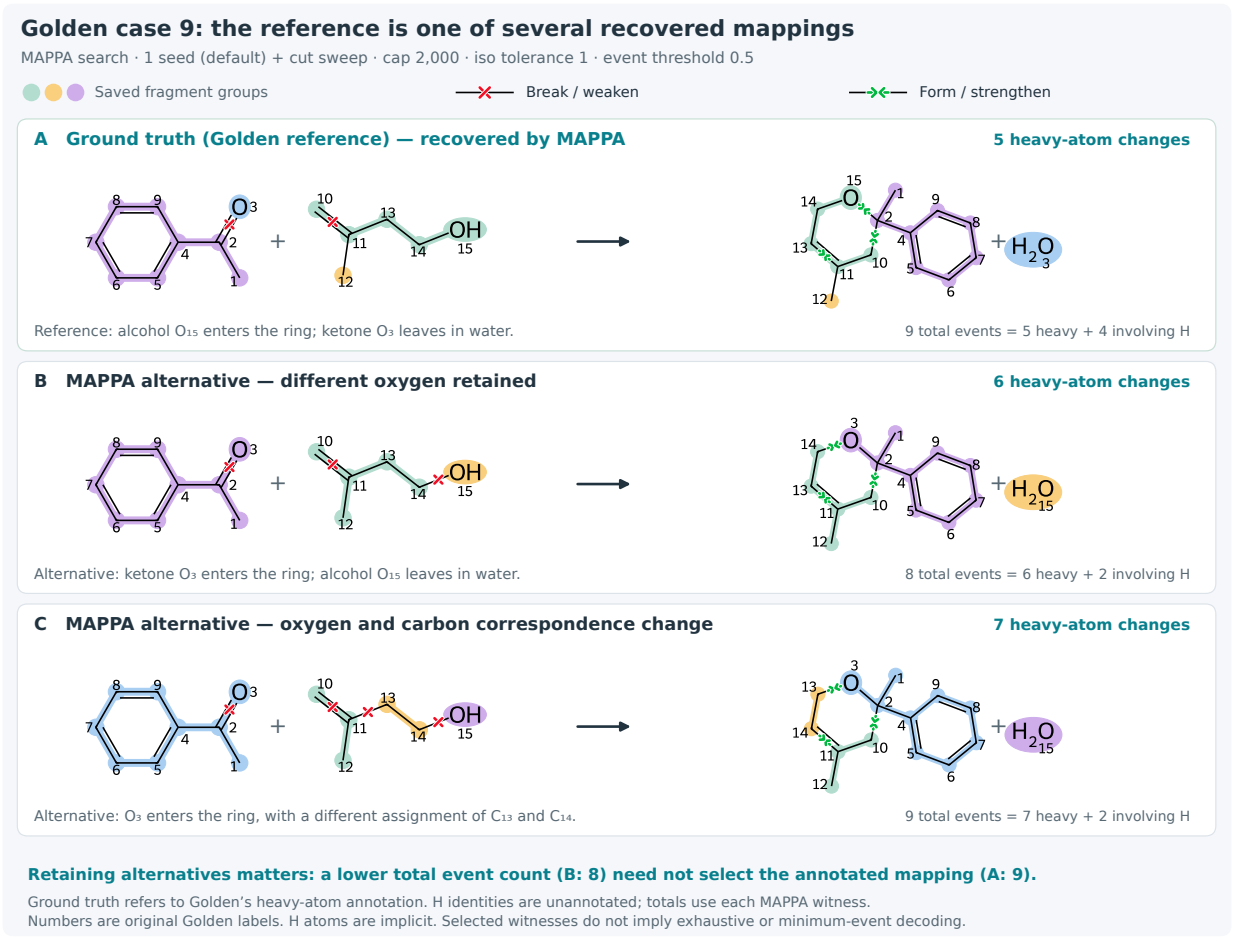


Figure 4 A reference correspondence and two alternatives for Golden case 9. A matches all 15 annotated heavy-atom pairs. B exchanges the oxygen origins; C also exchanges C_{13}/C_{14}. Colors identify matched fragments. Black lines show endpoint bonds, red × marks show cleavage or weakening, and green inward arrows show formation or strengthening. H atoms are implicit in the drawing but included in event totals; each witness minimizes H-involving changes for its fixed heavy-atom correspondence. "Ground truth" denotes the dataset annotation. These archived witnesses were obtained without reference labels using one-seed sweep search, cap 2,000 and $\tau_{\text{iso}} = 1$; events use threshold 0.5.

4 Discussion

MAPPA separates the decision needed during growth—whether one match exists—from the complete compressed representation needed when a fragment is committed. The final catalogue preserves structural alternatives and their correlated assignments, while local decoding reuses bond-response calculations across shared fragments. This makes correspondence search and chemical interpretation independently inspectable.

The method retains several limits. Exact fragment queries can still require substantial search on difficult inputs. Greedy growth, sampled seed orders, single-edge cuts and branch caps restrict structural coverage. Lossless representation of retained choices therefore does not imply exhaustive discovery of all mappings. Similarly, sharing local responses does not guarantee small tables for every family. Chemical conclusions depend on the input attributes, bond-weight model and event thresholds; a compatible correspondence is not evidence of a feasible mechanism.

Retained alternatives can support reaction-path preparation. Atom-index assignment affects endpoint interpolation (3), and different correspondences or joint symmetry choices could supply starting structures for nudged elastic band calculations (7). Local bond events can also identify alternative reaction cores for

mechanistic assessment. These are downstream uses of the representation; the present evaluation does not measure improvements in transition-state searches or pathway prediction.

The completed Golden study establishes reference-family recovery under its stated protocol. Broader annotated datasets and matched performance comparisons are needed to assess generalization and scaling. Historical studies remain available with their original scope in the supplement; they are not substituted for measurements of the present implementation.

Code and data

Code and reproduction instructions are available at MAPPA source repository, branch `main`. The repository guide `manuscript/REPRODUCIBILITY.md` links full benchmark configurations, software versions, per-case results, and figure-generation scripts. Golden is distributed by its original authors (9).

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A Technical algorithm details

A.1 Exact existence queries during growth

An existence query takes the growing source fragment F, its accepted preservation edges A, and the assignments locked by completed fragments. Its output is one compatible injective assignment, or proof that none exists. It does not construct the completed fragment's alternative placements.

For each source atom u, a domain D_u contains compatible target atoms. A locked atom has its recorded image as its only destination; other atoms exclude destinations occupied by completed fragments. Target degree must accommodate the accepted edges incident to u. For each accepted edge (u, v), a support relation contains the target pairs satisfying the graph floor and matching tolerance. These are constraints on accepted edges only, without additional nonedge or induced-subgraph tests.

Propagation repeatedly removes occupied singleton destinations from other domains and removes any destination lacking support in a neighboring domain. An empty domain or conflicting singletons proves infeasibility. A bipartite matching check then tests whether the remaining domains admit distinct destinations for every source atom; it can reject competing domains without enumerating their assignments. These checks are necessary conditions. If choices remain, the solver selects the smallest domain, breaking ties by accepted-edge degree and atom index, and explores its destinations until the first solution is found. A negative answer requires exhaustion of the surviving choices. Difficult negative instances can therefore still require search.

A growth proposal first tries to reuse the previous witness: it already satisfies the earlier edges, so only the proposed edge and any newly joined locked atoms require checking. If that extension fails, earlier assignments within the growing fragment may change in a new exact query. Completed-fragment locks remain fixed. An interrupted query does not establish absence. The Python and native implementations follow these same rules; the native propagation tests bond support by scanning target neighbors against indexed domain membership.

A.2 Direct compressed completion

Completion starts from the finished fragment constraints. It partitions eligible target atoms into interchangeable classes: swapping two atoms within a class must preserve their eligibility, the relevant accepted-edge support relations, and the orbit, lock and policy context. For custom atom policies, compatibility with future source atoms is retained as well. Local interchangeability is a property of these matching predicates. Component labels alone do not split an otherwise interchangeable free-atom class.

The second equivalence is the exact global target automorphism action. Its colored graph retains the target's exact bond weights, edge presence, atom policy and component identities. Locks and current allocation roles further restrict this action. It can move atoms outside the fragment together with atoms inside it; individual orbit labels cannot replace these correlated moves. The automorphisms are obtained with nauty (10).

The completion search allocates source atoms to target classes, rather than successively listing concrete bijections. After each allocation, the same support and injectivity propagation removes impossible continuations. Forced class choices are assigned directly. For an unresolved source atom, only one class representative from each orbit of the conditioned global action is explored. Conditioning preserves the allocations already made. A completed allocation yields a representative assignment, target pools and fixed roles; every represented injection within those pools satisfies the finished constraints. Pool allocations related by the allowed global action remain compressed. Thus distinct placements are constructed directly, without a subsequent candidate-merging pass.

A.3 Conditional branches, caps and retained histories

Each completed placement produces a conditional continuation for later fragments. The cap counts compressed choices, not the bijections inside them, and applies both to an individual expansion and to the combined live continuations across parents. Exact continuation states may share a node while retaining all incoming histories. After these joins, a parent's new descendants are admitted together or rejected together if their addition exceeds the cap; descendants already joined to retained states remain represented. A capped expansion is not a partial list of the first K placements and does not prove absence of other mappings.

Algorithm A1 Separate fragment growth and completion

Require: A seed, active source edges, completed-fragment locks and matching rules
Ensure: Completed compressed placements and accepted/deferred edges
Initialize the growing fragment and one witness; return no placement if none exists
Initialize the accepted preservation edges
while an eligible frontier proposal remains **do**
  Select a boundary atom and then an incident edge, each uniformly
  Form the proposed fragment, including a whole completed fragment if joined
  Retain the joined fragment's locked assignments and accepted preservation edges
  Extend the witness, or run the exact existence query with completed locks fixed
  if a witness exists **then**
    Accept the selected edge and replace the growing witness
  else if absence is proved **then**
    Defer the edge and continue with the remaining frontier
  end if
end while
Partition eligible target atoms by interchangeability for the finished constraints
Allocate source atoms to those classes using propagation and conditioned symmetry
Emit each distinct completed compressed placement once
Return placements to the reaction hierarchy for branching and cap checks

The search retains every history reaching a live continuation or returned result. Mapping and boundary payloads with no surviving descendant are released, while compact cap records retain their context, attempted count, limit and number of occurrences. This storage sharing does not merge distinct mapping relations. Neither the branch cap nor the cache budget bounds every temporary allocation or total process memory.

A.4 Finalization and conservative family reduction

Finalization supplies the saved symmetry actions needed to interpret each retained path. Family reduction then collects its terminal representative, required preservation edges, matching policy and ordered pool/group action program with earlier locks applied. It normalizes independent actions by commuting only those with disjoint support, and collapses identical adjacent full actions. Other overlapping actions retain their order.

Identical normalized records share one family. The public branch grouping uses the unordered collection of paired fragment atom sets, preserving the union of its family records, source histories and alternative fragment partitions. Equal witnesses, atom sets or automorphism groups alone do not establish family equality. This is conservative reduction of representations, not a semantic normal form proving that all remaining mapping sets are different or disjoint. It preserves the saved mapping union; it does not recover alternatives excluded by growth or caps.

A.5 Local event responses and valid joint composition

The decoder first completes an inventory of fragment atom-set pairs and every use in the catalogue's families and alternative partitions. This shares structural descriptions. Shared event responses additionally require agreement of the local witness, relevant ordered actions and boundary context. Local action positions are bound back to the original family positions, so one action moving several fragments remains one joint choice.

For each relevant source pair, decoding follows its reachable target endpoint pairs under the ordered action program. Each group closes its pair orbit before the next action acts. For two distinct endpoints in a pool of size k, this involves at most $k(k-1)$ ordered pairs rather than $k!$ permutations. Before computing that orbit, the decoder tests a superset formed from the individual destination domains: if every pair has the same event response, that response is constant on the allowed choices too. A constant contribution avoids pair expansion but does not remove the underlying action or its constraints.

The local tables include source bonds, pairs that can reach a target bond, inter-fragment relationships and mapped/unmatched boundaries. Responses use the original endpoint weights and strict event thresholds.

Algorithm A2 Prepare local responses and evaluate a requested family choice

Require: A reduced catalogue and original endpoint matrices
Inventory all unique fragments, occurrences, partitions, actions and boundaries
Build shared internal-fragment and boundary response tables
Retain the family bindings and constraints connecting those tables
for all requested joint choices **do**
    Validate the saved ordered actions, injectivity and preservation constraints
    Choose one saved fragment partition of the family
    Combine its internal and boundary responses, counting each bond once
    Return event counts and, when requested, the corresponding bond changes
end for

Target-side boundary contributions depend on which target atoms are occupied, without assigning correspondences to unmatched atoms. Constraints involving other atoms can restrict a pair projection, so the decoder validates the complete requested joint choice before combining local contributions. Each bond contributes once. Alternative partitions provide alternative decompositions of a family; their totals are not added.

This preparation neither rematches fragments nor enumerates complete bijections. Pair tables can still grow with the reachable domains. Ordinary decoding also does not enumerate unique whole-reaction event classes or choose a minimum-event mapping; the archived event-window study below used a separate explicit query.

A.6 Completed-fragment reuse

The cache stores completed compressed fragment results and the dependencies read during growth and completion. A reuse request must agree on the endpoint problem, matching rules, seed/random settings, branch settings and target occupancy. It then checks the assignments and completed-fragment memberships actually consulted, topology reads affected by changed cuts, and relevant deferred boundaries. Changes outside those dependencies can reuse the result; a changed relevant lock cannot. Unaffected external deferred edges are carried forward with the reused result.

Retention is bounded and oversized entries need not be cached. Eviction changes computation, not the returned placements or cap decisions. Shared snapshots carry endpoint, policy and code identity with integrity checks. At task boundaries a worker can load the latest compatible snapshot; an immutable first snapshot remains available if the latest stored version cannot be loaded. Publication replaces complete snapshots atomically. This is recovery of reusable data, not a different matching algorithm. Scheduling and storage do not alter the scientific search conditions.

B Reproducibility

The repository's `manuscript/REPRODUCIBILITY.md` links the benchmark scripts, pinned software and input versions, full configurations, and per-case records. It distinguishes the completed Golden validation in Section 3.1 from the archived September evaluations below. Table A1 gives the archived comparison settings. The guide `docs/PYTHON_API.md` demonstrates matching, family reduction, separate local decoding and conditional symmetry queries through the Python API.

The current Golden validation used source `096a010961f53944ddd4ddc433bd8d56f282a2f2`, with fresh results and caches. The current main-text timing figure sums total task-worker CPU, including computation and result publication, across both directions and all four stages, then divides by 1,851 reactions. Its mean is 10.01 CPU seconds per reaction. Coordinator activity, startup and idle capacity are excluded; this is not elapsed latency. Compute-only CPU is recorded separately and excludes publication. The current evidence does not record a CPU model, so these measurements do not establish a hardware-normalized comparison. They are not pooled with the historical timing cohorts.

The WBO-generation method for the archived coordinate inputs is not recorded in the available metadata, so regeneration of those matrices from molecular structures is not established.

The archived paired search timings use an Apple M4 Pro CPU. Averages use 1,821 common completed reactions. Reference verification and interrupted work are excluded. MAPPA includes checkpoint I/O; SLAP excludes I/O. Higher-seed runs measured on different CPUs are excluded from this paired comparison. For the archived coordinate stage comparison, MAPPA search includes checkpoint I/O; SLAP mapping excludes input preparation and export, and its hydrogen refinement includes I/O. Shared event rescoring and failed or replaced attempts are excluded. Archived Golden MAPPA timing includes symmetry finalization; SLAP includes graph construction, mapping, and export. In that primary Golden evaluation, uncut and swept SLAP have one and three unresolved cases, respectively; all MAPPA verdicts are resolved. These timing cohorts are not pooled with the current-design validation.

C Archived September evaluations

The following studies used their recorded September implementations. They retain the original numerical results and comparison scopes; they are not reruns of the current algorithm or direct performance comparisons with Section 3.1.

C.1 Archived datasets and comparator protocol

Golden inputs use formal bond orders and explicit H, with reference labels removed before search. Recovery requires the annotated heavy-atom correspondence, including unmatched atoms, modulo endpoint chemical symmetries preserving charge, isotope, H count, and stereochemistry. These chemical colors define reference equivalence; they do not imply that element/WBO matching itself enforces every chemical attribute. Failures and unresolved cases remain in the denominator. This measures reference inclusion among returned alternatives, not their ranking.

LocalMapper provides the prior accuracy-SOTA baseline (1). Its published Golden accuracy was 89.8% on 1,758 retained reactions with manual calibration. SLAP reported 86.9% on that dataset version, counting reference inclusion among its candidates (8). These literature scores are separate from our strict re-evaluation. We additionally apply the cut sweep introduced here to SLAP; "SLAP sweep" denotes this extension, not a published SLAP score. It is our strongest evaluated prior-method baseline for alternative recovery. Released RXNMapper, Chython, Indigo, and RDT provide further comparisons (13; 2; 5; 12).

The coordinate collection contains 18–137 explicit atoms per endpoint, including organometallic and multicomponent structures. It was inspected during development and has no verified atom mappings: it is not a mapping-accuracy benchmark. The archived study compared the lowest event count found and the patterns achieving that count. MAPPA used supplied WBO matrices; SLAP used XYZ-derived binary graphs. Returned SLAP heavy-atom assignments received hydrogen refinement and were scored on the same WBO matrices as MAPPA. The comparison covers those saved witnesses, not all possible SLAP hydrogen assignments or a fresh current-implementation evaluation.

Released comparators received the same label-free Golden inputs, without prediction repair or reference-guided ranking. Invalid endpoint chemistry, duplicate atom labels and other failures remained in the denominator. Binary SLAP was retained as the stronger default configuration; bidirectional comparisons used both binary and weighted modes. SLAP returns multiple candidates, whereas the other comparators return at most one explicit bijection. These results describe the evaluated interfaces and scoring rule, including its difference from RDT's chemical-change criterion. The repository records released models, versions, adapter settings and failure frequencies.

C.2 Archived evaluation configurations

Table A1 lists the archived settings. Search did not use reference or comparator correspondences. Full configurations, implementation versions, and reproduction scripts are provided in the repository (Appendix B).

Table A1 Archived evaluation configurations. Both use $b = 0.2$, $\tau_{iso} = 1$, explicit H, and uncut plus single-edge searches. One seed per cut and the sweep define the default setting. Archived Golden measurements cover sweep search; no-sweep controls retain only the uncut graph. A seed setting counts source-atom orderings per cut.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Bidirectional	Reactant to product
Seed orderings	1 (default), 2, 3, 10	1 (default)
Branch cap	100 (default)	100 (default), 2,000
Evaluation	Reference-family recovery	Signed event-class coverage

Table A2 Golden reference recovery and runtime in the archived evaluation. Recovery uses all 1,851 primary-evaluation records. MAPPA retains compressed families; SLAP returns explicit candidates; other methods return one bijection over their matched atoms. MAPPA uses cap 100 and bidirectional search. Bidirectional SLAP combines both directions and both binary/weighted modes; the default SLAP row uses binary only. Times are mean CPU seconds per reaction, excluding separate bond-event decoding. Unmarked times use the same 1,403 completed reactions on an AMD EPYC 9J14. *a*: archived completed calls (1,718–1,851 per method), CPU model unrecorded; these times are not directly comparable with unmarked times. †: LocalMapper, prior accuracy SOTA (2024). ‡: SLAP with our sweep, strongest evaluated prior-method recovery baseline.

Method	Search setting	Output	Reference recovered	Mean CPU s
MAPPA	1 seed, sweep (default)	Families	1,834 (99.08%)	0.963
MAPPA	2 seeds, sweep	Families	1,834 (99.08%)	1.619
MAPPA	3 seeds, sweep	Families	1,837 (99.24%)	2.265
MAPPA	10 seeds, sweep	Families	1,840 (99.41%)	6.993
MAPPA	1 seed, no sweep	Families	1,489 (80.44%)	0.040
SLAP, bidirectional	No sweep	Candidates	1,661 (89.74%)	0.105
SLAP, bidirectional‡	Sweep	Candidates	1,796 (97.03%)	2.793
SLAP, binary	Default, no sweep	Candidates	1,591 (85.95%)	0.075[a]
RXNMapper	Default, no sweep	One bijection	1,547 (83.58%)	0.056[a]
LocalMapper†	Default, no sweep	One bijection	1,605 (86.71%)	0.294[a]
Chython	Default, no sweep	One bijection	1,561 (84.33%)	0.612[a]
Indigo	Default, no sweep	One bijection	692 (37.39%)	0.184[a]
RDT	Default, no sweep	One bijection	1,030 (55.65%)	1.021[a]

C.3 Archived Golden reference recovery and search cost

In the archived campaign, the one-seed MAPPA configuration recovered 99.08% of Golden references, compared with 97.03% for SLAP sweep (Table A2 and figure A1). These scores belong to that campaign, separate from Section 3.1. The paired comparison includes 43 references recovered only by MAPPA and 5 only by SLAP. LocalMapper is the strongest evaluated method returning one bijection. Additional MAPPA seeds yielded modest recovery gains at increasing search cost; two seeds did not improve recovery over one.

The paired archived controls attribute 345 additional reference recoveries to the MAPPA sweep and 135 to the SLAP sweep. Without the sweep, one-seed MAPPA had lower coverage than bidirectional SLAP. This supports varying the growth constraints to expose alternatives excluded by greedy fragment extension (Section 2.5); symmetry expansion alone cannot restore discarded placements. These differences are not measurements of a current no-sweep control.

Smaller-first grows fragments from the reaction side with fewer explicit atoms (including H) and matches into the other side; larger-first reverses these roles. In the archived comparison, combining both directions added 31 recoveries over smaller-first alone, demonstrating complementary alternatives (Table A3).

Table A3 Archived recovery and runtime by search orientation. All rows use the sweep. Recovery uses all 1,851 Golden records; CPU statistics are seconds per reaction over the same 1,821 completed reactions (Apple M4 Pro CPU). Smaller-first grows from the side with fewer explicit atoms, including H; larger-first reverses the source and target. Equal-size ties use stored order and its reverse. Bidirectional output combines both searches, and its CPU cost is their sum. SLAP combines binary and weighted modes. CPU/SLAP compares means within each orientation. Timing scopes are specified in Appendix B.

Method	Orientation	Recovered	Mean	Median	95th pct.	CPU/SLAP
MAPPA, 1 seed (default)	Smaller-first	1,803 (97.41%)	1.141	0.263	5.289	0.34
	Larger-first	1,779 (96.11%)	1.377	0.348	6.401	0.92
	Bidirectional	1,834 (99.08%)	2.518	0.719	12.913	0.52
MAPPA, 2 seeds	Smaller-first	1,815 (98.06%)	2.232	0.416	10.635	0.67
	Larger-first	1,805 (97.51%)	2.721	0.578	14.064	1.83
	Bidirectional	1,834 (99.08%)	4.953	1.250	26.350	1.03
SLAP sweep (prior baseline)	Smaller-first	1,775 (95.89%)	3.329	0.422	4.752	1.00
	Larger-first	1,770 (95.62%)	1.490	0.455	4.568	1.00
	Bidirectional	1,796 (97.03%)	4.819	0.881	11.048	1.00

In these archived measurements, one-seed MAPPA had a lower mean and median search cost than SLAP sweep, but a higher 95th percentile. Two seeds approximately doubled the mean cost without improving recovery. These Golden timings cover search, excluding reference verification and separate bond-event decoding. The two timing tables use different experiments and should be compared within their stated CPU and reaction sets; neither establishes a current-implementation speed ranking.

In the archived cap study, increasing the branch cap from 100 to 2,000 yielded no additional swept recoveries among cases resolved at both caps, while increasing cost and unresolved outcomes at higher seed counts (Table A8). The uncut search gained 19 confirmed recoveries. Thus, in that study a larger branch budget did not substitute for changing the growth conditions.

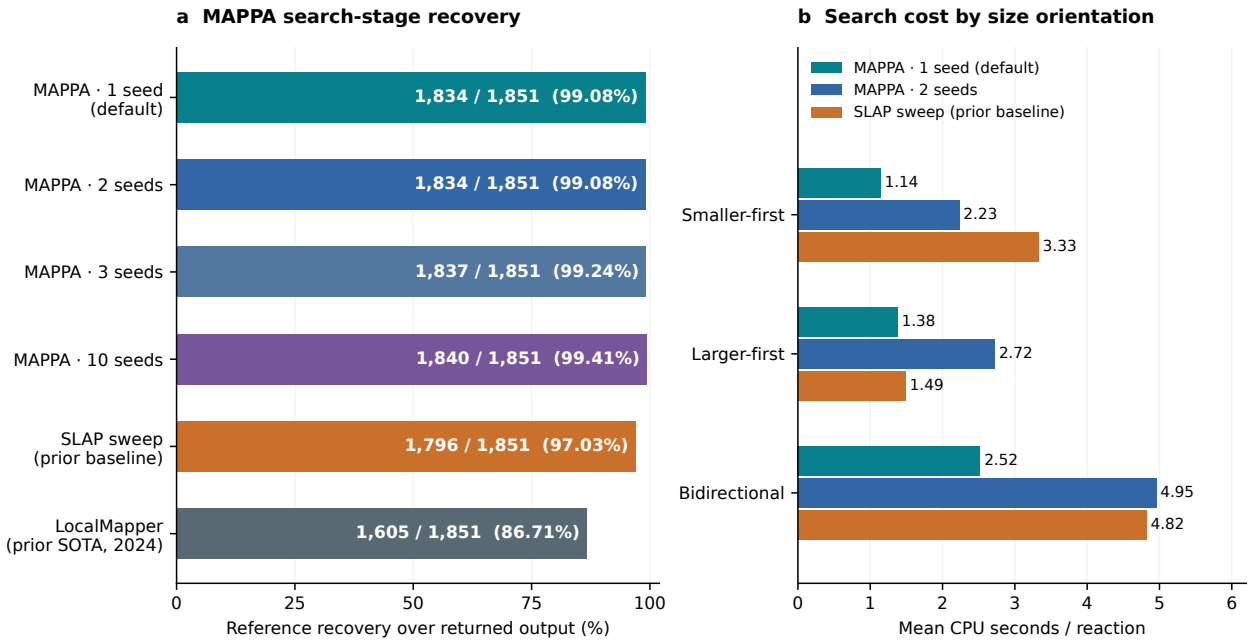


Figure A1 Archived Golden reference recovery and search cost. (a) Reference recovery over all 1,851 reactions. MAPPA and SLAP use our cut sweep; LocalMapper is the prior accuracy-SOTA baseline and returns one bijection. All scores use the archived evaluator. (b) Mean search CPU over the same 1,821 completed reactions, by orientation. Bond-event decoding is excluded. Full comparisons and timing scopes are in Tables A2 and A3.

C.4 Archived bond-event comparison on 140 coordinate reactions

The archived coordinate study used one-seed MAPPA sweep search with a branch cap of 2,000 and returned complete mappings for all 140 reactions. At cap 100, cases 123 and 125 returned no complete mapping. The 2,000-branch configuration therefore supplies this comparison, while 100 remains the library default. These completion counts and the event-window results below have not been remeasured with the current implementation.

The archived MAPPA output matched or lowered the minimum event count found by both native SLAP and SLAP sweep in every reaction (Table A4 and figure A2). This does not imply recovery of every alternative: at equal minima, six SLAP-sweep patterns remained absent across four reactions. An equal count can describe different changed bonds, and a lower count alone does not establish a more plausible transformation.

Table A4 Archived minimum bond-event comparison on 140 reactions. MAPPA uses one-seed sweep search at cap 2,000. The first three columns count reactions by MAPPA's minimum event count relative to the comparator. The final column gives comparator minimum-event patterns recovered by MAPPA among reactions with equal minima. Counts refer to returned candidates, not proven global optima.

Compared with	MAPPA fewer	Equal	MAPPA more	Patterns at equal minima
SLAP sweep	4	136	0	158/164
Native SLAP	15	125	0	138/140

To compare alternatives above MAPPA's own minima, the archived study queried through one event above the higher of the two SLAP minima for each reaction. MAPPA retained 162 of 168 SLAP-sweep minimum-event patterns and 151 of 160 native-SLAP patterns within these windows. The output contains 300 distinct patterns, including 166 at MAPPA's own minima. Every saved family was exhausted or certified within its window; this establishes completeness for those retained families, not exhaustive structural search.

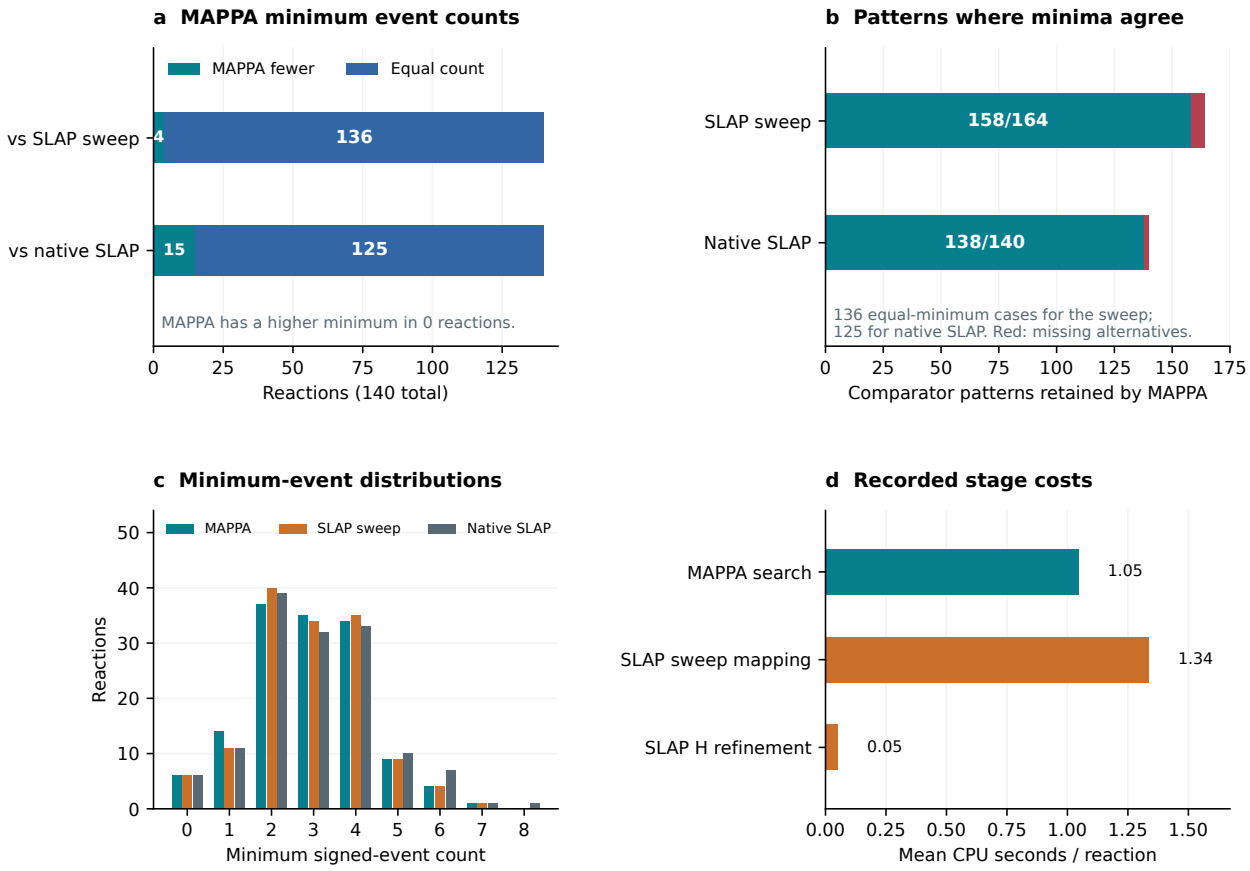


Figure A2 Archived bond-event comparison on 140 coordinate reactions. MAPPA uses one seed, the sweep, and cap 2,000. (a) Reactions with fewer or equal MAPPA events than each SLAP baseline. (b) Comparator minimum-event patterns recovered by MAPPA at equal minima. (c) Minimum-event count distributions. (d) Mean CPU for search and SLAP refinement, excluding MAPPA decoding (Table A5). Atom mappings are unverified; this compares bond events, not mapping accuracy.

Table A5 Archived stage CPU time per reaction on the 140 coordinate reactions. Measurements use an Apple M4 Pro CPU. MAPPA uses one-seed sweep search at cap 2,000. These stage costs exclude MAPPA decoding and do not establish an end-to-end speed ranking. SLAP refinement covers its returned witnesses. Timing boundaries are specified in Appendix B.

Recorded stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.047	0.314	3.481
SLAP sweep mapping	1.337	0.546	5.919
SLAP H refinement / initial scoring	0.053	0.011	0.207

The stage timings in Table A5 cover different outputs: MAPPA search retains families for subsequent decoding, while SLAP refinement operates on returned witnesses. They therefore do not establish an end-to-end speed ranking.

C.5 Archived event-window query scope

For each coordinate reaction, the archived event-window limit was one event above the larger of the native-SLAP and SLAP-sweep minima. All 120,052 retained families were exhausted within these limits, without a class-count cap. Higher-event patterns are outside this completeness claim. An event class records the signed

atom-pair changes, modulo source permutations preserving every thresholded bond-change test against the opposite endpoint. Each class has a concrete mapping witness. This explicit whole-reaction query is separate from ordinary decoding into shared local fragment responses.

Table A6 Archived complete event-window queries on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

The query used event invariance to certify constant families and encoded remaining choices in Z3 (4). A family was certified by exhaustion, a valid lower bound excluding the window, or unsatisfiability after excluding all discovered classes. An exhausted resource budget remained incomplete. The explicit query adapter can simplify adjacent nested action groups by exact generator-membership checks: if $H \leq G$, then $HG = GH = G$. This adapter simplification is separate from ordinary catalogue reduction in Appendix A.4. Completeness within an event window concerns the saved families, not exhaustive structural search.

C.6 Archived Golden nonrecoveries

Table A7 The eleven archived ten-seed Golden nonrecoveries. Case indices are zero-based. Retained pair counts cover every terminal in both directions and every saved cut at ten seed orderings and cap 100. The four cardinality mismatches concern exact reference relations, including unmatched atoms. This is not the nonrecovery set of the current one-seed validation.

Case	Reference pairs	Retained pairs	Observed mismatch
7	29	30	One extra matched pair
19	23	23	Atom pairing excluded by orbit bounds
590	31	31	Atom pairing excluded by orbit bounds
871	20	20	Atom pairing excluded by orbit bounds
986	16	16	Atom pairing excluded by orbit bounds
1228	12	12	Atom pairing excluded by orbit bounds
1285	14	14	Joint C/O correspondence excluded
1358	36	37	One extra matched pair
1377	22	23	One extra matched pair
1475	18	19	One extra matched pair
1553	20	20	Atom pairing excluded by orbit bounds

Independent reconstruction agrees with the supplied references in all eleven cases. Four require a different number of matched pairs; seven require correspondences absent from the retained families. All have at least one capped direction, but this does not identify the cap as the cause. The archived uncapped traces below demonstrate greedy-growth exclusions in two cases. Reference-directed feasibility checks are diagnostic and do not contribute to benchmark recovery. Per-case certificates and traces are linked in the repository.

Two archived traces show how greedy extension excluded a representable reference. One reference exchanges adjacent labels in an otherwise unchanged pyridine ring; another exchanges the carbonyl-oxygen attachments of oxalic acid across its retained carbons. Growth preserved bonds that these correspondences require separating. Uncapped trajectories still missed both references, while reference-constrained fragment choices established their representability. In the oxalic-acid example, a reference-selected three-edge relaxation also exposed the reference to matching; the single-edge sweep did not contain that cut set. These are feasibility diagnostics, not additional benchmark recoveries or traces of the current matcher. For the other five correspondence misses, the audit did not isolate cap truncation from unsampled growth choices. They motivate selective alternatives to successful growth extensions and explicit choices about leaving atoms unmatched, rather than treating larger caps or more symmetry expansion as sufficient remedies.

Only one of the eleven misses is recovered by any tested comparator configuration, including the SLAP sweep. The shared failures warrant inspection of the supplied correspondences as well as search improvements; they do not establish annotation errors. All eleven remain in the reported denominator. Independent evaluation on annotated coordinate reactions is needed to assess generalization beyond the development collection.

C.7 Archived branch-cap and CPU-time ablation on Golden

The archived study ran the twelve configurations in Table A8 on all 1,851 Golden reactions using an AMD EPYC 9J14 CPU. Both search directions are unioned. SLAP unions its binary and weighted modes. Search and reference verification have separate fixed resource budgets. The recorded cap-100 recovery counts reproduce the archived primary evaluation. At cap 2,000, unresolved counts increase with seed count. Among reactions resolved at both caps, no swept configuration gains or loses a reference recovery.

Table A8 Archived Golden branch-cap ablation with paired SLAP baselines. Recovery counts confirmed reference inclusion over all 1,851 records; unknown cases remain in the denominator. CPU values are seconds per reaction. Paired CPU uses the same 1,403 reactions whose searches finish under every configuration. Spent CPU averages recorded search cost over all 1,851 selected attempts, including interrupted calls; it is not completed-search latency. Both timings exclude reference verification and initial imports. Superseded attempts are excluded. Cap 100 with a one-seed sweep is the default setting.

Method	Cap	Seeds	Recovered (%)	Unknown	Paired CPU	Spent CPU
MAPPA uncut	100	1	1,489 (80.44)	0	0.040	0.085
MAPPA sweep (default)	100	1	1,834 (99.08)	0	0.963	3.873
MAPPA sweep	100	2	1,834 (99.08)	0	1.619	7.260
MAPPA sweep	100	3	1,837 (99.24)	0	2.265	10.556
MAPPA sweep	100	10	1,840 (99.41)	0	6.993	33.211
MAPPA uncut	2,000	1	1,508 (81.47)	5	0.086	0.517
MAPPA sweep	2,000	1	1,834 (99.08)	2	4.317	30.055
MAPPA sweep	2,000	2	1,834 (99.08)	4	8.283	48.549
MAPPA sweep	2,000	3	1,837 (99.24)	6	12.201	61.890
MAPPA sweep	2,000	10	1,835 (99.14)	12	43.843	113.777
SLAP uncut	–	–	1,661 (89.74)	1	0.105	0.659
SLAP sweep	–	–	1,794 (96.92)	5	2.793	9.895

The ten-seed cap-2,000 run leaves five references recovered at cap 100 unresolved, accounting for its lower confirmed count. The bounded SLAP sweep confirms 1,794 references here, compared with 1,796 in the archived primary evaluation: two previously recovered cases are unresolved under this study's limits. These outcomes are reported separately from verified exclusions. The paired timing cohort excludes 448 reactions because at least one configuration does not finish; the all-attempt cost column exposes the additional computational cost on those difficult reactions. Source pins, individual outcomes, directional costs, and reproduction scripts are provided with the code.

C.8 Archived end-to-end timing on coordinate reactions

The archived September implementation completed search and event-window decoding for 140/140 coordinate reactions. On this full set, the full pipeline takes a mean 6.07 CPU seconds per reaction (median 0.81; 95th percentile 25.47). Decoding accounts for a mean 4.65 CPU seconds (median 0.29). These measurements include search, final-family construction, decoding, and candidate output; they are separate from the search-only comparison above (Table A9).

The historical study measured one-seed, cap-2,000 cut-sweep search followed by its event-window decoder on the Apple M4 Pro CPU. CPU times are reported per reaction. The event thresholds and windows match the bond-event comparison, with no class-count cap. All 140 completed reactions reproduce the archived event-class sets and family counts. Median end-to-end wall time is 0.96 seconds. Timing includes input loading, checkpoint I/O, final-family construction, decoding, and candidate serialization; initial imports and

reference-set verification are excluded. All 140 reactions finish within the five-minute watchdog. Detailed timing records and reproduction instructions are provided with the code.

Table A9 Archived MAPPA search plus event-window queries. CPU seconds per completed reaction in the 140-reaction pass. The final row measures the whole pipeline; medians and percentiles of individual stages are not additive. Completion coverage and timing scope are stated above.

Stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.257	0.300	3.468
Final-family catalogue	0.156	0.019	0.595
Bond-event decoding	4.647	0.288	24.361
Candidate serialization	0.005	0.001	0.015
End to end	6.073	0.812	25.470